Course Name:	Web Development Laboratory (316U01L306)	**Semester:**	**III**
Date of Performance:	**15/09/2026**	**DIV/ Batch No:**	**A2**
Student Name:	**Kehan Desai**	**Roll No:**	**16010125050**

Experiment No: 5

Title: JavaScript Programming for Climate Intelligence – Heatwave Monitoring, Prediction, and Early Warning

Aim of the Experiment: To develop JavaScript programs for a Climate Intelligence system that monitors heatwave conditions, analyzes environmental information, determines heatwave risk, and provides early-warning recommendations using JavaScript concepts such as variables, operators, expressions, control statements, objects, functions, arrays, strings, and regular express

Objectives for the Experiment:

- To use JavaScript variables, values, operators, and expressions for processing climate-related information.
- To apply control statements for determining heatwave risk and generating appropriate warnings.
- To create JavaScript objects representing climate monitoring information.
- To use functions for performing climate-related calculations and generating recommendations.
- To use arrays for storing and analyzing temperature and environmental readings.
- To use strings and regular expressions for validating location and user information.
- To integrate JavaScript with HTML forms for a Climate Intelligence early-warning application.

COs to be achieved:

CO1: Developing webpages using HTML, CSS and JavaScript.

Books/ Journals/ Websites references:

1. MDN Web Docs – JavaScript Guide and HTML Forms.
2. W3Schools – JavaScript.

Theory:

Task 1: Heatwave Temperature Analysis

Problem Statement

Develop a JavaScript program to accept the current temperature and threshold temperature of a location and determine the heatwave condition.

Functional Requirements

Declare variables for:

- Location name
- Current temperature
- Heatwave threshold temperature
- Humidity
- Monitoring date

Perform the following operations:

1. Calculate the difference between the current temperature and heatwave threshold.
2. Determine whether the current temperature has crossed the threshold.
3. Display the heatwave monitoring status.
4. Display the location and temperature information.
5. Use JavaScript message-printing methods such as:
    - alert()
    - document.write()
    - console.log()

Expected Output

Location: Mumbai
Current Temperature: 39°C
Heatwave Threshold: 37°C
Temperature Difference: 2°C

Status: Heatwave Condition Detected

Concepts Tested

- Values
- Variables
- Data types
- Operators

- Expressions
- Arithmetic operations
- Message printing
- Basic conditional statement

Deliverable

heatwave_analysis.html

Task 2: Heatwave Risk and Early-Warning Decision

Problem Statement

Develop a JavaScript program that recommends the appropriate heatwave warning level based on temperature, humidity, and forecast conditions.

Input Parameters

- Current temperature
- Humidity
- Forecast temperature
- Heatwave condition
- Vulnerable population alert requirement

Decision Rules

Condition	Recommendation
Temperature ≥ 40°C	**Severe Heatwave Warning**
Temperature ≥ 38°C and Humidity ≥ 60%	**High Heat Risk – Issue Warning**
Temperature between 35°C and 38°C	**Moderate Heat Risk – Monitor Conditions**
Forecast temperature ≥ 40°C	**Early Warning – Prepare for Heatwave**
Temperature < 35°C	**Normal – Continue Monitoring**

Requirements

1. Accept climate values using JavaScript variables.
2. Use if, else-if, and else statements.
3. Use relational operators such as >, <, >=, and <=.
4. Use logical operators such as && and ||.
5. Display the appropriate heatwave risk level.
6. Display an early-warning recommendation.

Example Output

Location: Pune
Temperature: 39°C
Humidity: 65%

Risk Level: HIGH
Recommendation: Issue Heatwave Warning

Concepts Tested

- Variables
- Operators
- Relational operators
- Logical operators
- Expressions
- if-else-if
- Control statements
- Message printing

Deliverable

heatwave_warning.html

Task 3: Create a JavaScript Object for Climate Monitoring

Problem Statement

Create a JavaScript object representing a location monitored by the Climate Intelligence system.

Object Properties

The climate monitoring object should contain:

- Location name
- City
- Temperature
- Humidity
- Wind speed
- Forecast temperature
- Heatwave risk level
- Alert status

- Monitoring date

Required Methods

Create methods to:

1. Display complete climate information.
2. Calculate the temperature difference from the threshold.
3. Determine the heatwave risk level.
4. Display the early-warning status.
5. Update the alert status.

Example Object

```
let climateData = {
   location: "Mumbai",
   temperature: 39,
   humidity: 65,
   windSpeed: 12,
   forecastTemperature: 41,
   riskLevel: "High",
   alertStatus: "Active",

   displayData: function() {
      console.log("Location: " + this.location);
      console.log("Temperature: " + this.temperature + "°C");
      console.log("Humidity: " + this.humidity + "%");
      console.log("Risk Level: " + this.riskLevel);
   }
};
```

Requirements

1. Create the object using object-literal syntax.
2. Access object properties using the dot operator.
3. Create functions/methods inside the object.
4. Use this to access object properties.
5. Display the climate information on the webpage or console.

Concepts Tested

- Object-oriented JavaScript
- Objects
- Properties
- Methods

- Functions
- this keyword
- Conditional statements

Deliverable

climate_object.html

Task 4: Analyze Temperature Readings Using Arrays

Problem Statement

A Climate Intelligence system collects temperature readings at regular intervals. Store the readings in a JavaScript array and analyze the temperature conditions.

Consider the following temperature readings:

let temperatureReadings = [34, 36, 38, 39, 41, 40, 37, 35, 39, 42];

Perform the Following Operations

1. Display all temperature readings.
2. Find the minimum temperature.
3. Find the maximum temperature.
4. Calculate the average temperature.
5. Count the number of readings above the heatwave threshold.
6. Count the number of readings equal to or above 40°C.
7. Display all readings that indicate a high heat condition.

Example Output

Temperature Readings:
34 36 38 39 41 40 37 35 39 42

Minimum Temperature: 34°C
Maximum Temperature: 42°C
Average Temperature: 38.1°C
Readings ≥ 40°C: 3
Heatwave Condition Readings: 41, 40, 42

Concepts Tested

- Arrays

- Array traversal
- Loops
- Array methods
- Functions
- Conditional statements
- Mathematical calculations

Deliverable

temperature_analysis.html

Task 5: Validate Climate Monitoring Registration Details

Problem Statement

Develop a JavaScript program to validate the details entered by a user for registering with the Climate Intelligence Heatwave Early-Warning System.

Fields to Validate

- User name
- Mobile number
- Email address
- Location name
- User ID
- PIN/Postal Code

Validation Rules

Field	Validation Rule
User Name	Only alphabets and spaces
Mobile Number	Exactly 10 digits
Email	Valid email format
Location	Alphabets and spaces
User ID	Format: CLM-1234
PIN Code	Exactly 6 digits

Example

User Name: Rahul Sharma
Mobile Number: 9876543210
Email: rahul@gmail.com

Location: Mumbai
User ID: CLM-1234
PIN Code: 400077

Requirements

1. Accept user details using JavaScript.
2. Validate the user name.
3. Validate the mobile number.
4. Validate the email address.
5. Validate the location.
6. Validate the Climate Intelligence User ID.
7. Display appropriate error messages.
8. Display a success message when all fields are valid.

Regular Expression Examples

User Name:

/^[A-Za-z]+$/

Mobile Number:

/^[6-9][0-9]{9}$/

Climate User ID:

/^CLM-[0-9]{4}$/

PIN Code:

/^[0-9]{6}$/

Concepts Tested

- Strings
- String methods
- Regular expressions
- test()
- Functions
- Boolean expressions
- Conditional statements

Deliverable

climate_validation.html

Task 6: Develop a Climate Intelligence Registration and Early-Warning Form

Problem Statement

Develop an HTML-based registration form for users who want to access the **Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning System**.

The form should collect user, location, and alert-preference information and validate the entered data using JavaScript.

Input Fields

Create a registration form containing:

- User Name
- Mobile Number
- Email
- User ID
- Location/City
- PIN/Postal Code
- Age Group
- User Category
- Preferred Alert Channel
- Registration Date

Suggested User Categories

- General Public
- Senior Citizen
- Student
- Outdoor Worker
- Healthcare Worker
- Government/Disaster Management
- Other

Validation Requirements

1. No mandatory field should remain empty.
2. User name must contain only alphabets and spaces.
3. Mobile number must contain exactly 10 digits.
4. Email address must follow a valid format.

5. User ID must follow the pattern:

CLM-1234

6. PIN/Postal Code must contain exactly six digits.
7. User Category must be selected.
8. Preferred Alert Channel must be selected.
9. Location/City must not be empty.
10. Registration date must be entered.
11. Registration date should not be a future date.
12. Appropriate error messages should be displayed beside invalid fields.
13. Invalid fields should be highlighted.
14. The form should not be submitted until all validation rules are satisfied.
15. After successful validation, display:

Registration Successful!

You are registered for Climate Intelligence
Heatwave Monitoring and Early-Warning Alerts.

HTML Elements to Use

Students should use:

- <form>
- <label>
- <input>
- <select>
- <option>
- <fieldset>
- <legend>
- <button>

JavaScript Requirements

The program should use JavaScript to:

1. Access HTML form elements using the DOM.
2. Read values entered by the user.
3. Validate the input values.
4. Use functions for individual validation operations.
5. Use regular expressions for structured fields.

6. Use conditional statements.
7. Handle the form submit event.
8. Display validation errors.
9. Prevent invalid form submission.
10. Display a success message after valid registration.

Code:-

Task 1:-

```
function analyzeHeatwave() {
    const location = document.getElementById("analysisLocation").value.trim();
    const currentTemperature = Number(
        document.getElementById("analysisCurrentTemp").value
    );
    const thresholdTemperature = Number(
        document.getElementById("analysisThreshold").value
    );
    const humidity = Number(
        document.getElementById("analysisHumidity").value
    );
    const monitoringDate =
        document.getElementById("analysisDate").value;

    const temperatureDifference =
        currentTemperature - thresholdTemperature;

    let status;

    if (currentTemperature >= thresholdTemperature) {
        status = "Heatwave Condition Detected";
    } else {
        status = "No Heatwave Condition";
    }

    document.getElementById("analysisResult").innerHTML =
        "Location: " + location + "<br>" +
        "Current Temperature: " + currentTemperature + "°C<br>" +
        "Heatwave Threshold: " + thresholdTemperature + "°C<br>" +
        "Temperature Difference: " + temperatureDifference + "°C<br>" +
        "Humidity: " + humidity + "%<br>" +
        "Monitoring Date: " + monitoringDate + "<br>" +
        "Status: " + status;
```

```
    console.log("Location: " + location);
    console.log("Current Temperature: " + currentTemperature + "°C");
    console.log("Heatwave Threshold: " + thresholdTemperature + "°C");
    console.log("Temperature Difference: " + temperatureDifference + "°C");
    console.log("Status: " + status);

    alert(status);
}

function displayHeatwaveReport() {
    const location = document.getElementById("analysisLocation").value;
    const currentTemperature = Number(
        document.getElementById("analysisCurrentTemp").value
    );
    const thresholdTemperature = Number(
        document.getElementById("analysisThreshold").value
    );

    const difference =
        currentTemperature - thresholdTemperature;

    const status =
        currentTemperature >= thresholdTemperature
            ? "Heatwave Condition Detected"
            : "No Heatwave Condition";

    const outputWindow = window.open("", "_blank");

    outputWindow.document.write(
        "<h2>Heatwave Monitoring Status</h2>" +
        "<p>Location: " + location + "</p>" +
        "<p>Current Temperature: " + currentTemperature + "°C</p>" +
        "<p>Heatwave Threshold: " + thresholdTemperature + "°C</p>" +
        "<p>Temperature Difference: " + difference + "°C</p>" +
        "<p>Status: " + status + "</p>"
    );
}

Task 2:-
function determineHeatwaveRisk() {
    const location =
        document.getElementById("riskLocation").value.trim();

    const temperature = Number(
        document.getElementById("riskTemp").value
```

```
);

const humidity = Number(
  document.getElementById("riskHumidity").value
);

const forecastTemperature = Number(
  document.getElementById("riskForecast").value
);

const vulnerableAlert =
  document.getElementById("riskVulnerable").checked;

let riskLevel;
let recommendation;

if (temperature >= 40) {
  riskLevel = "SEVERE";
  recommendation = "Severe Heatwave Warning";
} else if (
  temperature >= 38 &&
  humidity >= 60
) {
  riskLevel = "HIGH";
  recommendation = "Issue Heatwave Warning";
} else if (
  forecastTemperature >= 40
) {
  riskLevel = "EARLY WARNING";
  recommendation = "Prepare for Heatwave";
} else if (
  temperature >= 35 &&
  temperature < 38
) {
  riskLevel = "MODERATE";
  recommendation = "Monitor Conditions";
} else {
  riskLevel = "NORMAL";
  recommendation = "Continue Monitoring";
}

if (
  vulnerableAlert &&
  riskLevel !== "NORMAL"
) {
```

```
        recommendation +=
            " - Priority Alert for Vulnerable Population";
    }

    document.getElementById("riskResult").innerHTML =
        "Location: " + location + "<br>" +
        "Temperature: " + temperature + "°C<br>" +
        "Humidity: " + humidity + "%<br>" +
        "Forecast Temperature: " + forecastTemperature + "°C<br>" +
        "Risk Level: " + riskLevel + "<br>" +
        "Recommendation: " + recommendation;
}

Task 3:-
let climateData = {
    location: "Mumbai",
    city: "Mumbai",
    temperature: 39,
    humidity: 65,
    windSpeed: 12,
    forecastTemperature: 41,
    threshold: 37,
    riskLevel: "",
    alertStatus: "",
    monitoringDate: "2026-09-16",

    calculateTemperatureDifference: function () {
        return this.temperature - this.threshold;
    },

    determineRiskLevel: function () {
        if (this.temperature >= 40) {
            this.riskLevel = "Severe";
        } else if (
            this.temperature >= 38 &&
            this.humidity >= 60
        ) {
            this.riskLevel = "High";
        } else if (
            this.forecastTemperature >= 40
        ) {
            this.riskLevel = "Early Warning";
        } else if (
            this.temperature >= 35
        ) {
```

```
        this.riskLevel = "Moderate";
      } else {
        this.riskLevel = "Normal";
      }

      return this.riskLevel;
    },

    displayEarlyWarningStatus: function () {
      if (this.riskLevel === "Severe") {
        return "Severe Heatwave Warning";
      } else if (this.riskLevel === "High") {
        return "Issue Heatwave Warning";
      } else if (this.riskLevel === "Early Warning") {
        return "Prepare for Heatwave";
      } else if (this.riskLevel === "Moderate") {
        return "Monitor Conditions";
      } else {
        return "Continue Monitoring";
      }
    },

    updateAlertStatus: function () {
      if (this.riskLevel === "Normal") {
        this.alertStatus = "Inactive";
      } else {
        this.alertStatus = "Active";
      }
    },

    displayData: function () {
      return (
        "Location: " + this.location + "<br>" +
        "City: " + this.city + "<br>" +
        "Temperature: " + this.temperature + "°C<br>" +
        "Humidity: " + this.humidity + "%<br>" +
        "Wind Speed: " + this.windSpeed + " km/h<br>" +
        "Forecast Temperature: " +
        this.forecastTemperature + "°C<br>" +
        "Risk Level: " + this.riskLevel + "<br>" +
        "Alert Status: " + this.alertStatus + "<br>" +
        "Monitoring Date: " + this.monitoringDate
      );
    }
};
```

```
function displayClimateObject() {
    climateData.location =
        document.getElementById("objectLocation").value;

    climateData.city =
        document.getElementById("objectCity").value;

    climateData.temperature = Number(
        document.getElementById("objectTemp").value
    );

    climateData.humidity = Number(
        document.getElementById("objectHumidity").value
    );

    climateData.windSpeed = Number(
        document.getElementById("objectWind").value
    );

    climateData.forecastTemperature = Number(
        document.getElementById("objectForecast").value
    );

    climateData.threshold = Number(
        document.getElementById("objectThreshold").value
    );

    climateData.monitoringDate =
        document.getElementById("objectDate").value;

    climateData.determineRiskLevel();
    climateData.updateAlertStatus();

    const difference =
        climateData.calculateTemperatureDifference();

    const warning =
        climateData.displayEarlyWarningStatus();

    document.getElementById("objectResult").innerHTML =
        climateData.displayData() +
        "<br>Temperature Difference: " +
        difference +
        "°C<br>" +
```

```
        "Early Warning: " +
        warning;
}

Task 4:-
const temperatureReadings = [
    34, 36, 38, 39, 41,
    40, 37, 35, 39, 42
];

function analyzeTemperatureReadings() {
    const threshold = Number(
        document.getElementById("arrayThreshold").value
    );

    const minimumTemperature =
        Math.min(...temperatureReadings);

    const maximumTemperature =
        Math.max(...temperatureReadings);

    let total = 0;

    for (
        let i = 0;
        i < temperatureReadings.length;
        i++
    ) {
        total += temperatureReadings[i];
    }

    const averageTemperature =
        total / temperatureReadings.length;

    const aboveThreshold =
        temperatureReadings.filter(
            function (temperature) {
                return temperature > threshold;
            }
        );

    const atOrAbove40 =
        temperatureReadings.filter(
            function (temperature) {
                return temperature >= 40;
```

```
        }
      );

    document.getElementById("arrayReadings").innerHTML =
      temperatureReadings.join(", ");

    document.getElementById("arrayMin").textContent =
      minimumTemperature + "°C";

    document.getElementById("arrayMax").textContent =
      maximumTemperature + "°C";

    document.getElementById("arrayAvg").textContent =
      averageTemperature.toFixed(1) + "°C";

    document.getElementById("array40").textContent =
      atOrAbove40.length;

    document.getElementById("arrayAbove").textContent =
      aboveThreshold.length;

    document.getElementById("arrayHighHeat").textContent =
      atOrAbove40.join(", ");
}
```

Task 5:-

```
function validateClimateDetails(event) {
    event.preventDefault();

    const userName =
      document.getElementById("validateName").value.trim();

    const mobile =
      document.getElementById("validateMobile").value.trim();

    const email =
      document.getElementById("validateEmail").value.trim();

    const location =
      document.getElementById("validateLocation").value.trim();

    const userId =
      document.getElementById("validateUserId").value.trim();

    const pin =
```

```
    document.getElementById("validatePin").value.trim();

const namePattern =
    /^[A-Za-z ]+$/;

const mobilePattern =
    /^[6-9][0-9]{9}$/;

const emailPattern =
    /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;

const locationPattern =
    /^[A-Za-z ]+$/;

const userIdPattern =
    /^CLM-[0-9]{4}$/;

const pinPattern =
    /^[0-9]{6}$/;

let valid = true;

if (!namePattern.test(userName)) {
    document.getElementById(
        "validateNameError"
    ).textContent =
        "Enter a valid name.";

    valid = false;
} else {
    document.getElementById(
        "validateNameError"
    ).textContent = "";
}

if (!mobilePattern.test(mobile)) {
    document.getElementById(
        "validateMobileError"
    ).textContent =
        "Enter a valid 10-digit mobile number.";

    valid = false;
} else {
    document.getElementById(
        "validateMobileError"
```

```
    ).textContent = "";
}

if (!emailPattern.test(email)) {
    document.getElementById(
        "validateEmailError"
    ).textContent =
        "Enter a valid email address.";

    valid = false;
} else {
    document.getElementById(
        "validateEmailError"
    ).textContent = "";
}

if (!locationPattern.test(location)) {
    document.getElementById(
        "validateLocationError"
    ).textContent =
        "Enter a valid location.";

    valid = false;
} else {
    document.getElementById(
        "validateLocationError"
    ).textContent = "";
}

if (!userIdPattern.test(userId)) {
    document.getElementById(
        "validateUserIdError"
    ).textContent =
        "User ID must be in CLM-1234 format.";

    valid = false;
} else {
    document.getElementById(
        "validateUserIdError"
    ).textContent = "";
}

if (!pinPattern.test(pin)) {
    document.getElementById(
        "validatePinError"
```

```
      ).textContent =
         "PIN code must contain exactly 6 digits.";

      valid = false;
   } else {
      document.getElementById(
         "validatePinError"
      ).textContent = "";
   }

   if (valid) {
      document.getElementById(
         "validationResult"
      ).innerHTML =
         "Details verified successfully for Kehan Desai.";
   } else {
      document.getElementById(
         "validationResult"
      ).innerHTML =
         "Please correct the highlighted fields.";
   }

   return valid;
}

document.getElementById(
   "validationForm"
).addEventListener(
   "submit",
   validateClimateDetails
);

Task 6:-
function registerForAlerts(event) {
   event.preventDefault();

   const name =
      document.getElementById("registerName").value.trim();

   const mobile =
      document.getElementById("registerMobile").value.trim();

   const email =
      document.getElementById("registerEmail").value.trim();
```

```
const userId =
    document.getElementById("registerUserId").value.trim();

const location =
    document.getElementById("registerLocation").value.trim();

const pin =
    document.getElementById("registerPin").value.trim();

const ageGroup =
    document.getElementById("registerAgeGroup").value;

const category =
    document.getElementById("registerCategory").value;

const alertChannel =
    document.getElementById("registerAlertChannel").value;

const registrationDate =
    document.getElementById("registerDate").value;

const namePattern =
    /^[A-Za-z ]+$/;

const mobilePattern =
    /^[6-9][0-9]{9}$/;

const emailPattern =
    /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;

const userIdPattern =
    /^CLM-[0-9]{4}$/;

const locationPattern =
    /^[A-Za-z ]+$/;

const pinPattern =
    /^[0-9]{6}$/;

const today =
    new Date();

today.setHours(
    0,
    0,
```

```
    0,
    0
);

let validDate = false;

if (registrationDate !== "") {
    const selectedDate =
        new Date(
            registrationDate +
            "T00:00:00"
        );

    validDate =
        selectedDate <= today;
}

let valid = true;

if (!namePattern.test(name)) {
    document.getElementById(
        "registerNameError"
    ).textContent =
        "Enter a valid name.";

    valid = false;
} else {
    document.getElementById(
        "registerNameError"
    ).textContent = "";
}

if (!mobilePattern.test(mobile)) {
    document.getElementById(
        "registerMobileError"
    ).textContent =
        "Enter a valid 10-digit mobile number.";

    valid = false;
} else {
    document.getElementById(
        "registerMobileError"
    ).textContent = "";
}
```

```
if (!emailPattern.test(email)) {
    document.getElementById(
        "registerEmailError"
    ).textContent =
        "Enter a valid email address.";

    valid = false;
} else {
    document.getElementById(
        "registerEmailError"
    ).textContent = "";
}

if (!userIdPattern.test(userId)) {
    document.getElementById(
        "registerUserIdError"
    ).textContent =
        "User ID must be in CLM-1234 format.";

    valid = false;
} else {
    document.getElementById(
        "registerUserIdError"
    ).textContent = "";
}

if (!locationPattern.test(location)) {
    document.getElementById(
        "registerLocationError"
    ).textContent =
        "Enter a valid location.";

    valid = false;
} else {
    document.getElementById(
        "registerLocationError"
    ).textContent = "";
}

if (!pinPattern.test(pin)) {
    document.getElementById(
        "registerPinError"
    ).textContent =
        "PIN must contain exactly 6 digits.";
```

```
    valid = false;
} else {
    document.getElementById(
        "registerPinError"
    ).textContent = "";
}

if (ageGroup === "") {
    document.getElementById(
        "registerAgeGroupError"
    ).textContent =
        "Select an age group.";

    valid = false;
} else {
    document.getElementById(
        "registerAgeGroupError"
    ).textContent = "";
}

if (category === "") {
    document.getElementById(
        "registerCategoryError"
    ).textContent =
        "Select a user category.";

    valid = false;
} else {
    document.getElementById(
        "registerCategoryError"
    ).textContent = "";
}

if (alertChannel === "") {
    document.getElementById(
        "registerAlertChannelError"
    ).textContent =
        "Select an alert channel.";

    valid = false;
} else {
    document.getElementById(
        "registerAlertChannelError"
    ).textContent = "";
}
```

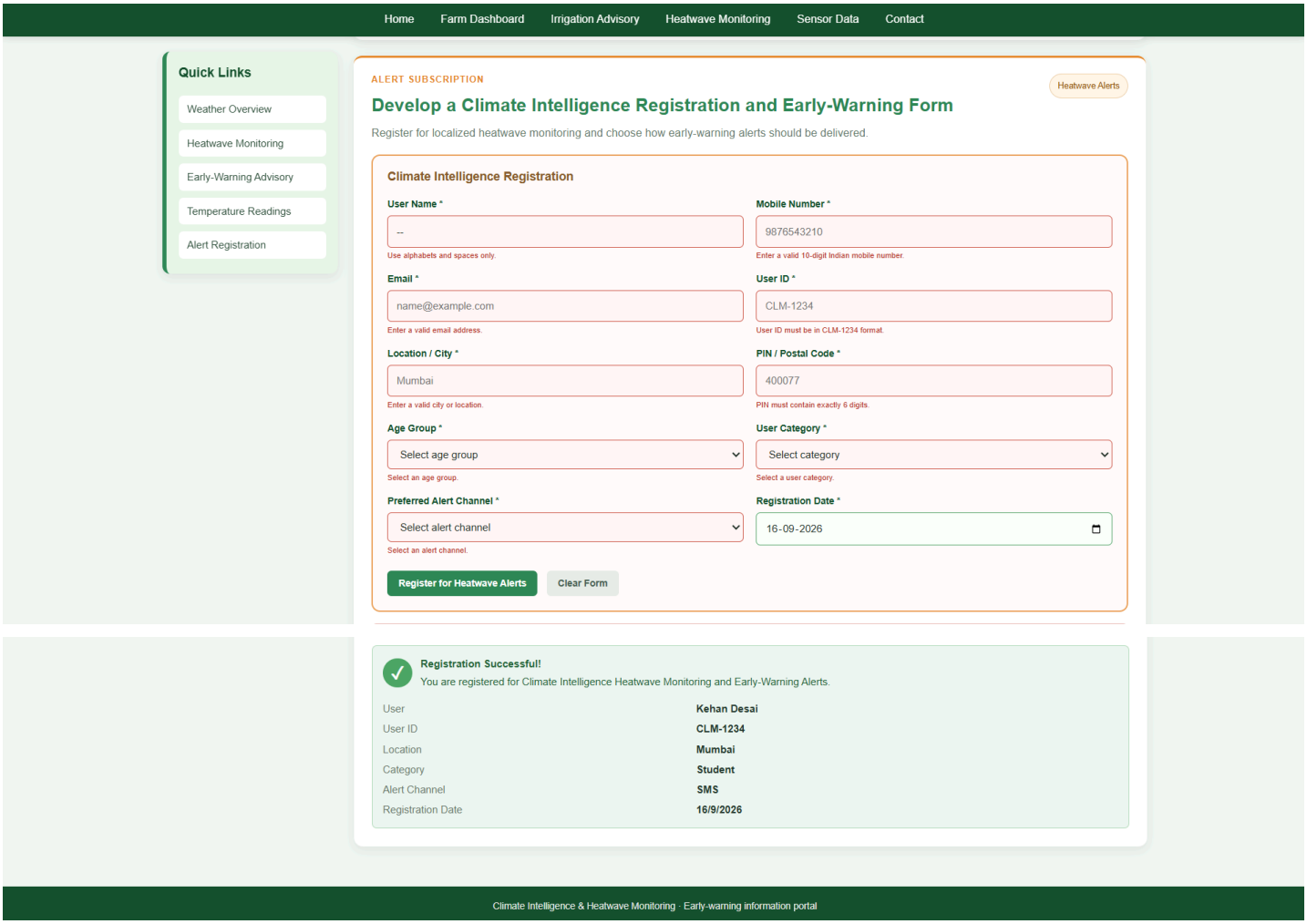
```
    if (!validDate) {
        document.getElementById(
            "registerDateError"
        ).textContent =
            "Registration date cannot be in the future.";

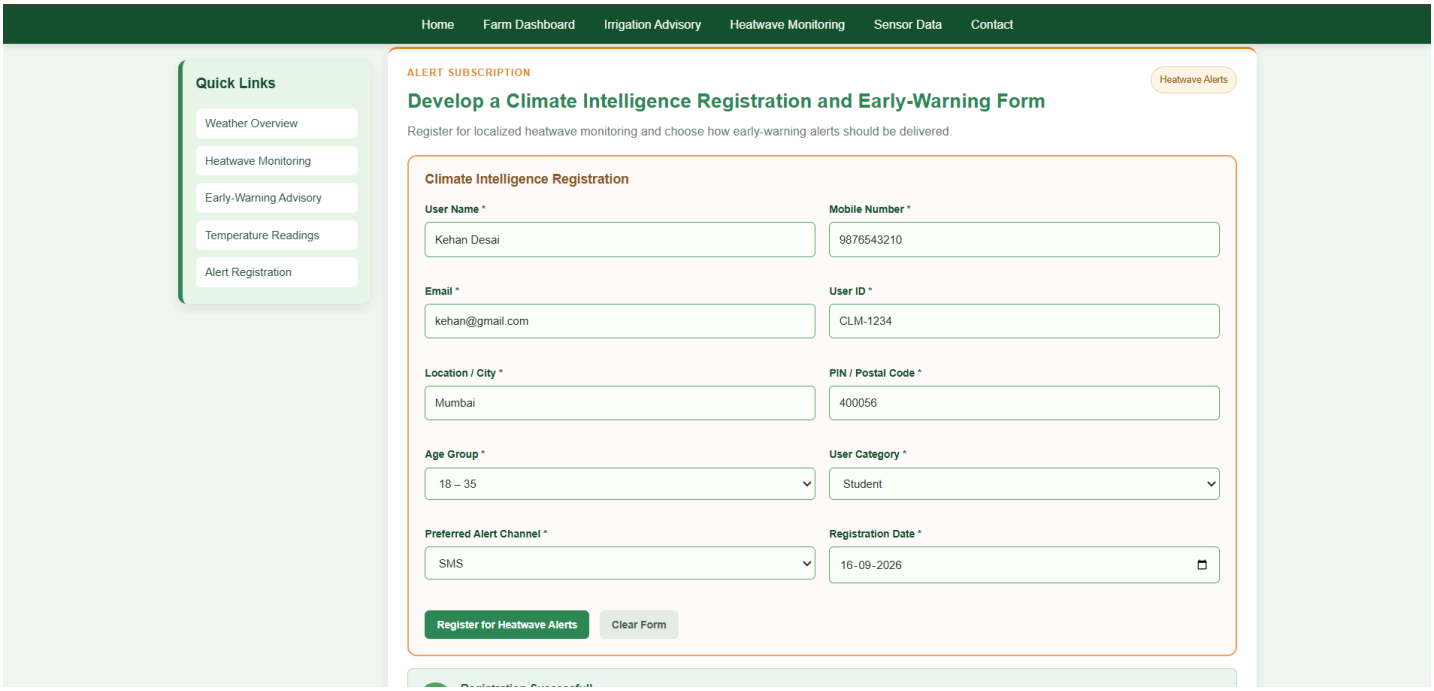
        valid = false;
    } else {
        document.getElementById(
            "registerDateError"
        ).textContent = "";
    }

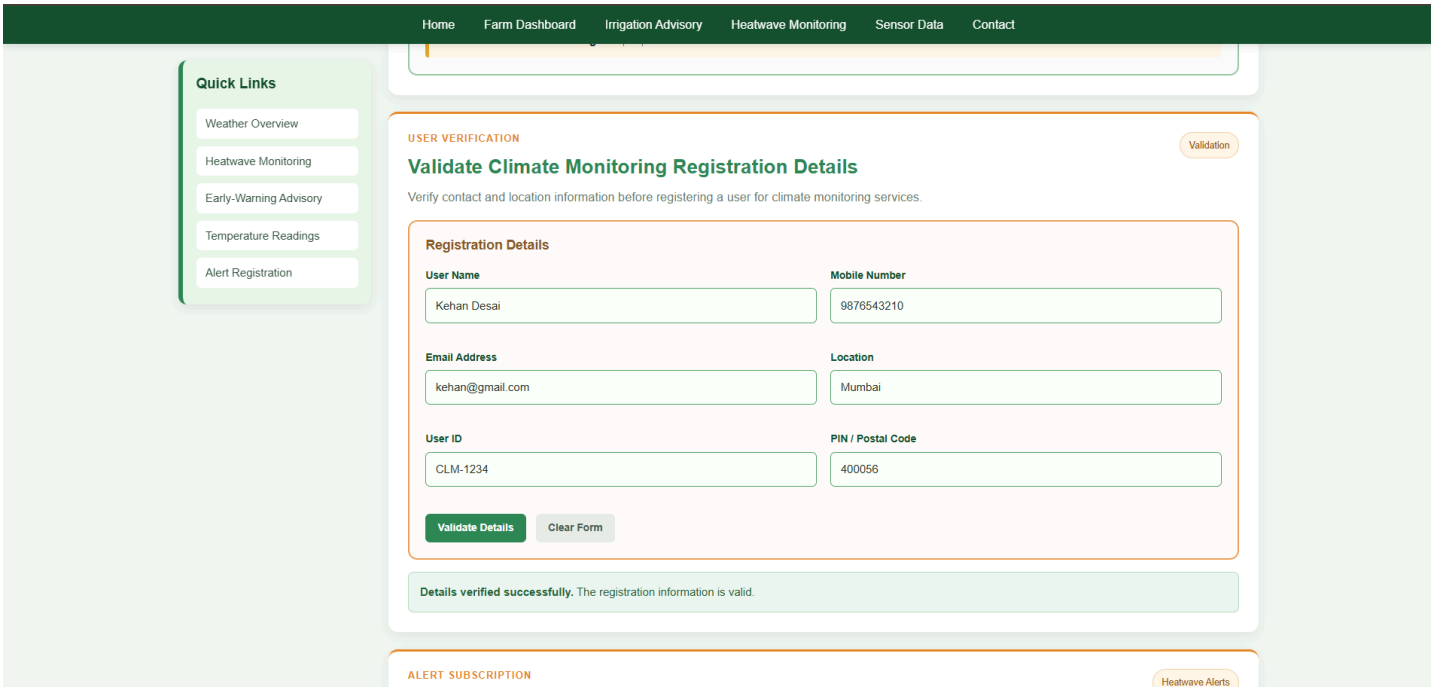
    if (valid) {
        document.getElementById(
            "registrationResult"
        ).innerHTML =
            "<strong>Registration Successful!</strong><br>" +
            "You are registered for Climate Intelligence " +
            "Heatwave Monitoring and Early-Warning Alerts.<br><br>" +
            "Name: " + name + "<br>" +
            "User ID: " + userId + "<br>" +
            "Location: " + location + "<br>" +
            "Preferred Alert Channel: " + alertChannel;
    } else {
        document.getElementById(
            "registrationResult"
        ).innerHTML =
            "Please correct the invalid details.";
    }

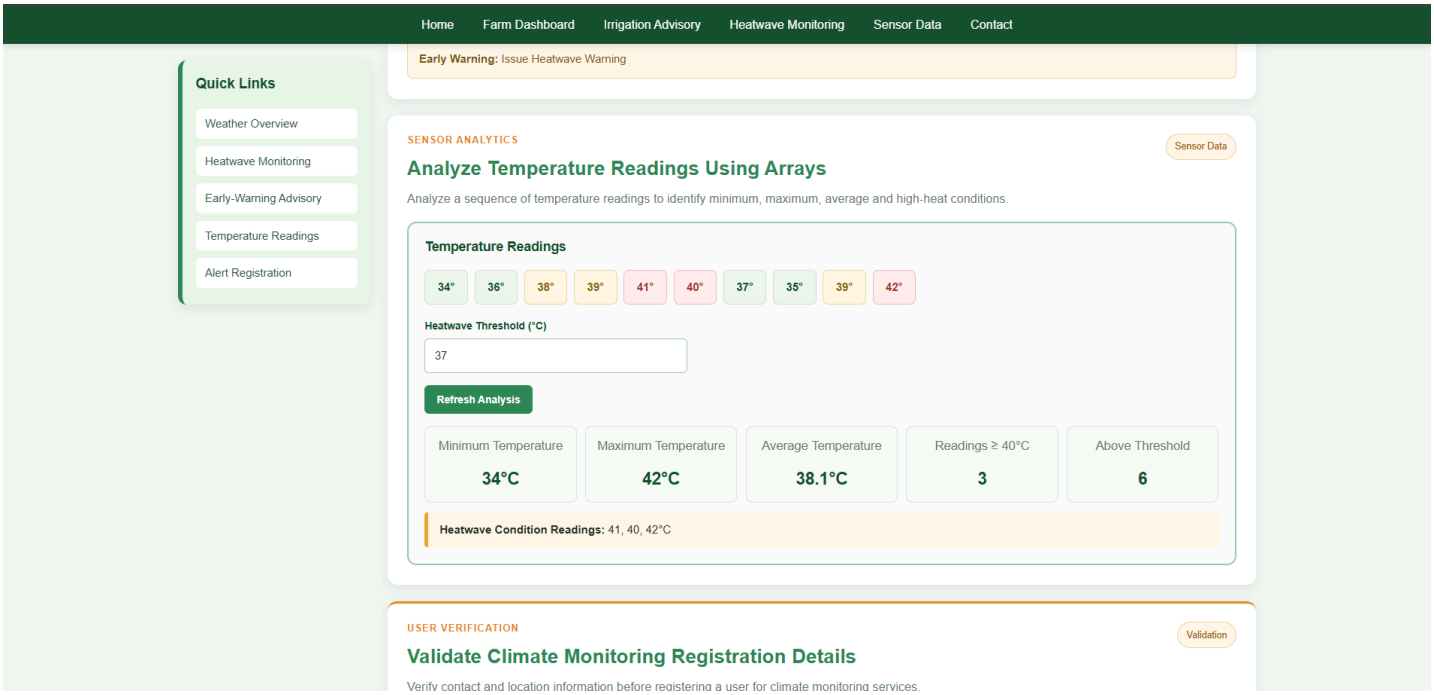
    return valid;
}

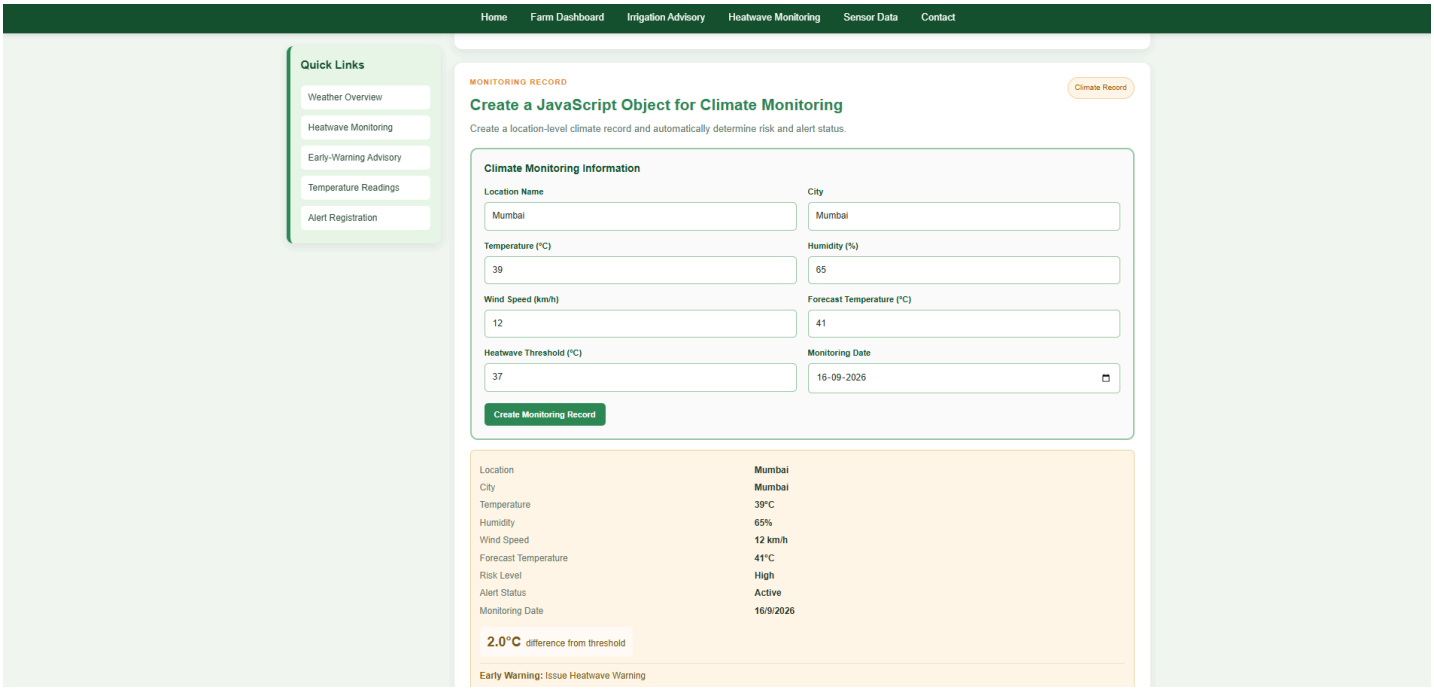
document.getElementById(
    "registrationForm"
).addEventListener(
    "submit",
    registerForAlerts
);
```

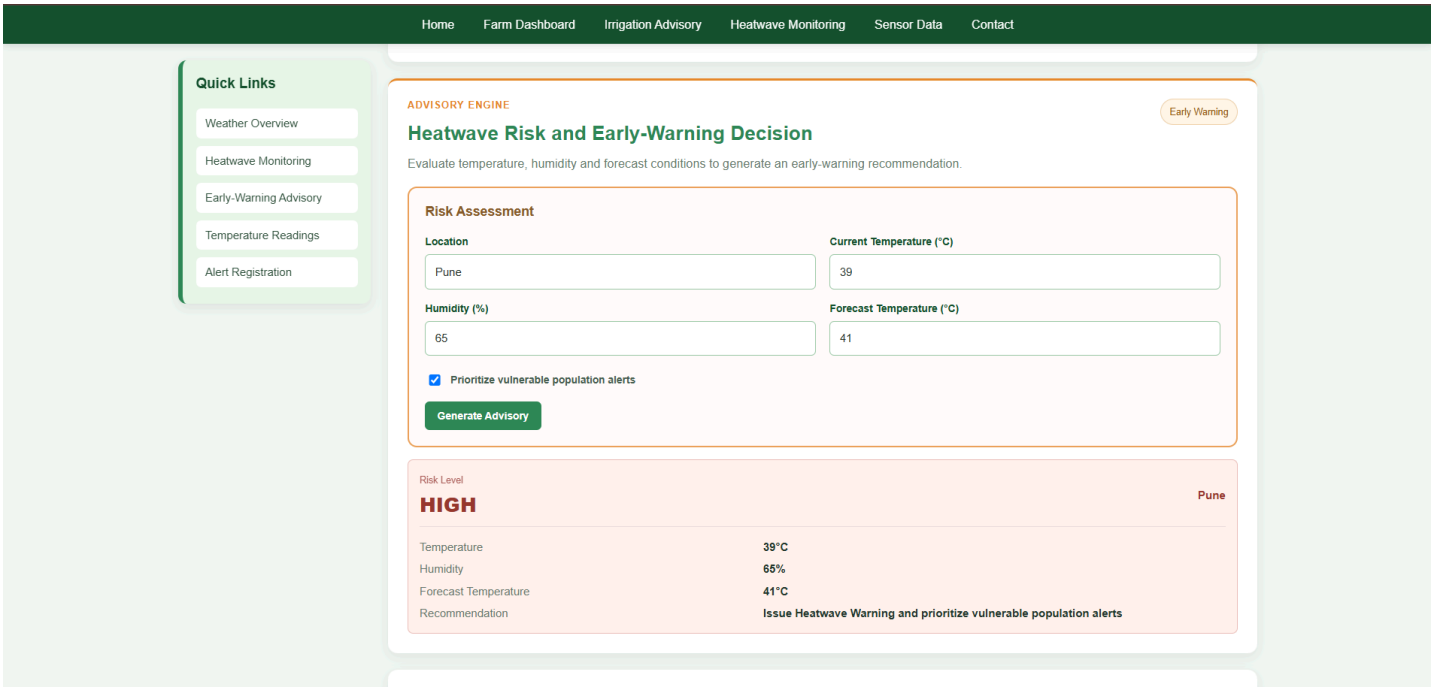

Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
ALERT SUBSCRIPTION
Heatwave Alerts
Develop a Climate Intelligence Registration and Early-Warning Form
Register for localized heatwave monitoring and choose how early-warning alerts should be delivered.
Climate Intelligence Registration
User Name *
--
Use alphabets and spaces only.
Mobile Number *
9876543210
Enter a valid 10-digit Indian mobile number.
Email *
name@example.com
Enter a valid email address.
User ID *
CLM-1234
User ID must be in CLM-1234 format.
Location / City *
Mumbai
Enter a valid city or location.
PIN / Postal Code *
400077
PIN must contain exactly 6 digits.
Age Group *
Select age group
Select an age group.
User Category *
Select category
Select a user category.
Preferred Alert Channel *
Select alert channel
Select an alert channel.
Registration Date *
16-09-2026
Register for Heatwave Alerts
Clear Form
Registration Successful!
You are registered for Climate Intelligence Heatwave Monitoring and Early-Warning Alerts.
User
Kehan Desai
User ID
CLM-1234
Location
Mumbai
Category
Student
Alert Channel
SMS
Registration Date
16/9/2026
Climate Intelligence & Heatwave Monitoring · Early-warning information portal

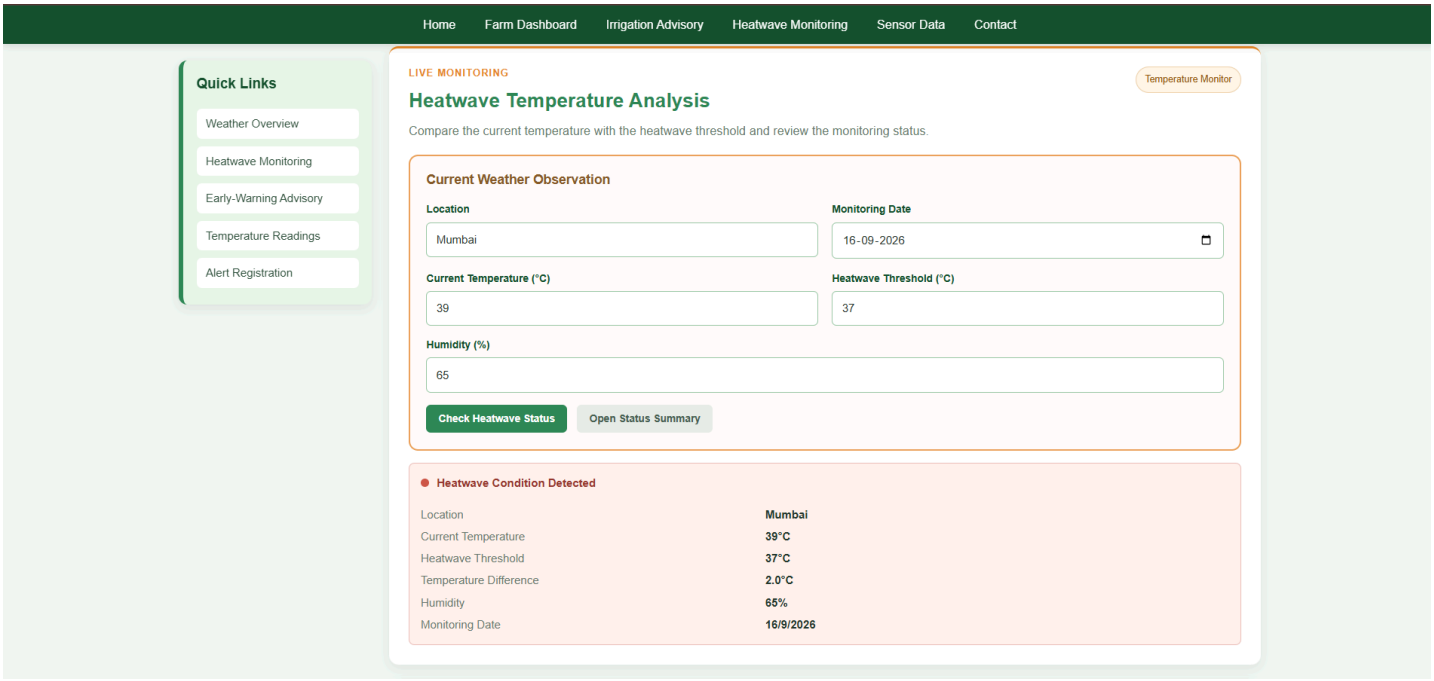

Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
ALERT SUBSCRIPTION
Heatwave Alerts
Develop a Climate Intelligence Registration and Early-Warning Form
Register for localized heatwave monitoring and choose how early-warning alerts should be delivered.
Climate Intelligence Registration
User Name *
Kehan Desai
Mobile Number *
9876543210
Email *
kehan@gmail.com
User ID *
CLM-1234
Location / City *
Mumbai
PIN / Postal Code *
400056
Age Group *
18 – 35
User Category *
Student
Preferred Alert Channel *
SMS
Registration Date *
16-09-2026
Register for Heatwave Alerts
Clear Form

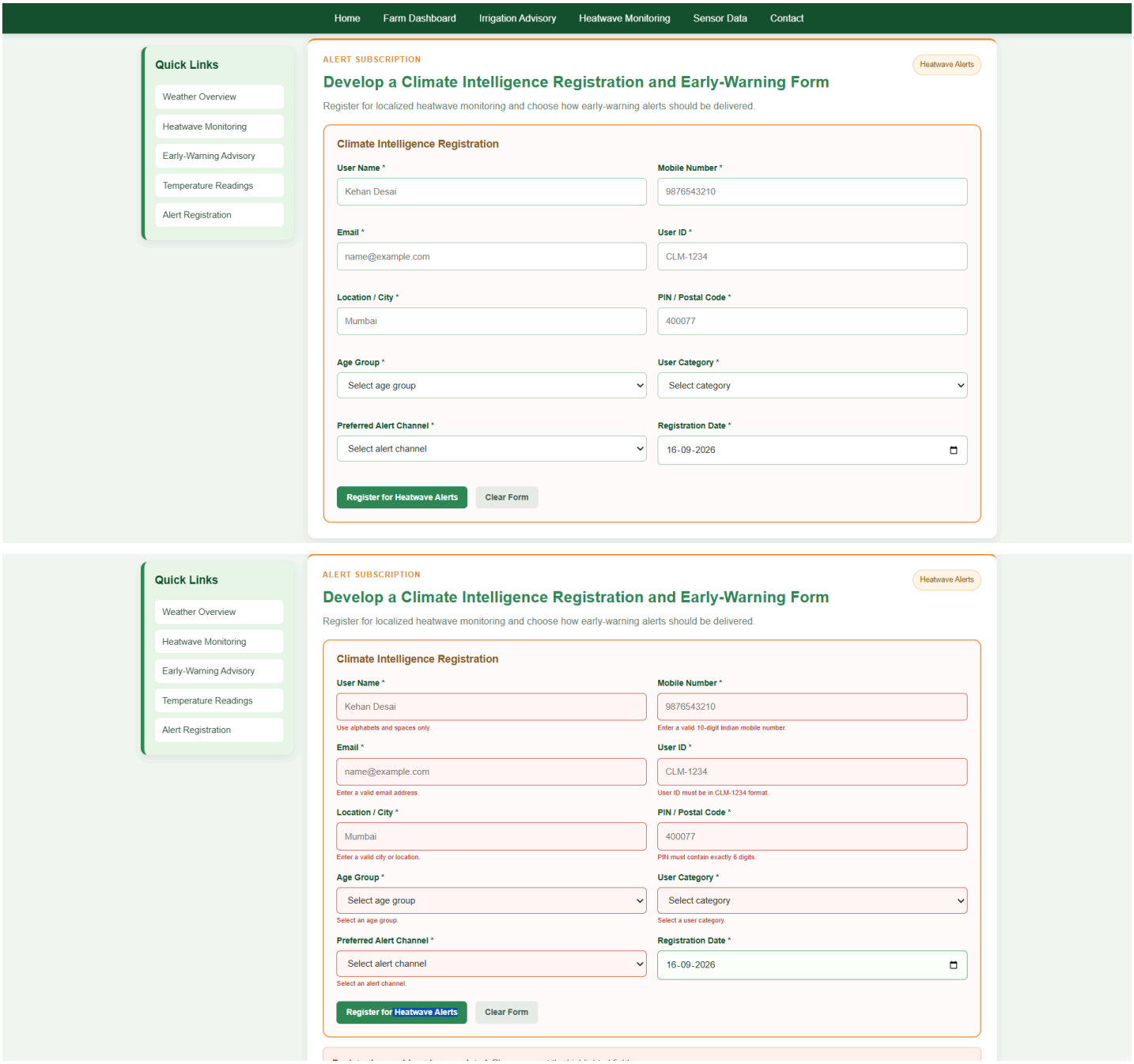

Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
USER VERIFICATION
Validation
Validate Climate Monitoring Registration Details
Verify contact and location information before registering a user for climate monitoring services.
Registration Details
User Name
Kehan Desai
Mobile Number
9876543210
Email Address
kehan@gmail.com
Location
Mumbai
User ID
CLM-1234
PIN / Postal Code
400056
Validate Details
Clear Form
Details verified successfully. The registration information is valid.
ALERT SUBSCRIPTION
Heatwave Alerts


Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Early Warning: Issue Heatwave Warning
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
SENSOR ANALYTICS
Sensor Data
Analyze Temperature Readings Using Arrays
Analyze a sequence of temperature readings to identify minimum, maximum, average and high-heat conditions.
Temperature Readings
34° 36° 38° 39° 41° 40° 37° 35° 39° 42°
Heatwave Threshold (°C)
37
Refresh Analysis
Minimum Temperature
34°C
Maximum Temperature
42°C
Average Temperature
38.1°C
Readings ≥ 40°C
3
Above Threshold
6
Heatwave Condition Readings: 41, 40, 42°C
USER VERIFICATION
Validation
Validate Climate Monitoring Registration Details
Verify contact and location information before registering a user for climate monitoring services.

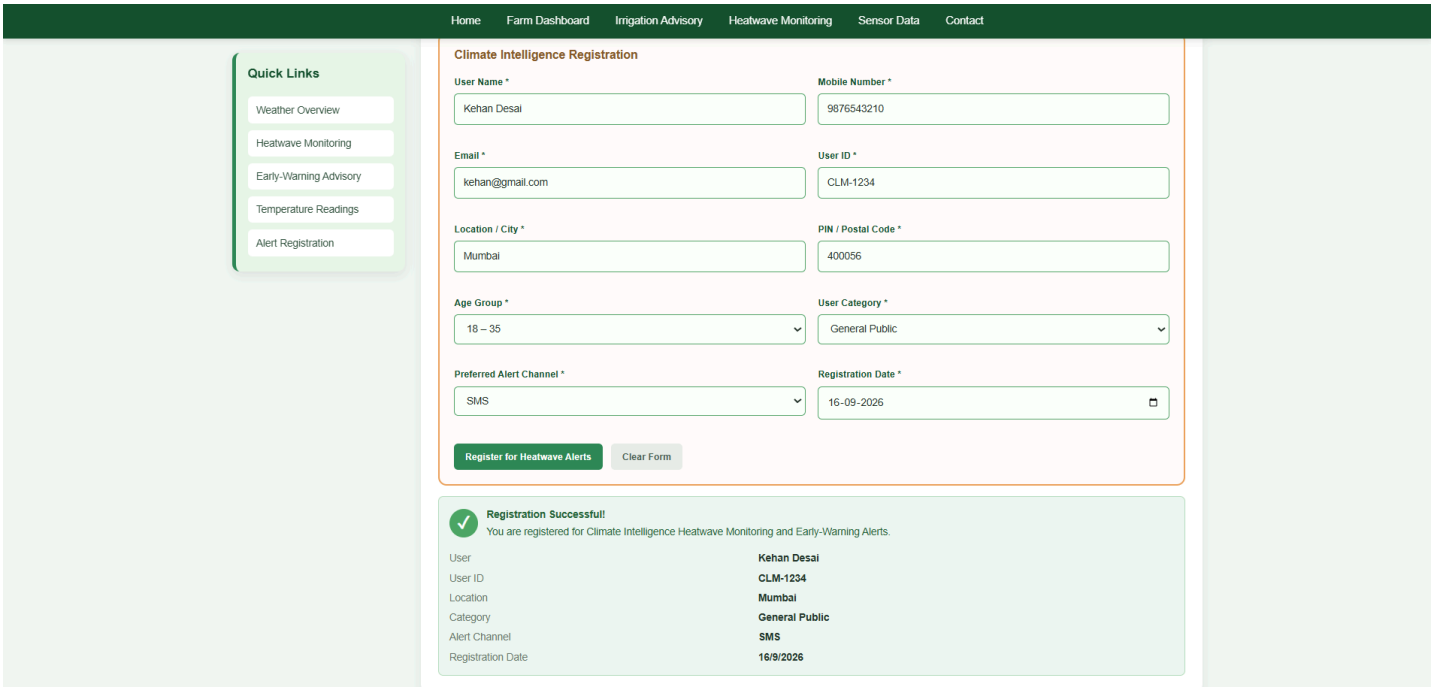

Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
MONITORING RECORD
Climate Record
Create a JavaScript Object for Climate Monitoring
Create a location-level climate record and automatically determine risk and alert status.
Climate Monitoring Information
Location Name
Mumbai
City
Mumbai
Temperature (°C)
39
Humidity (%)
65
Wind Speed (km/h)
12
Forecast Temperature (°C)
41
Heatwave Threshold (°C)
37
Monitoring Date
16-09-2026
Create Monitoring Record
Location Mumbai
City Mumbai
Temperature 39°C
Humidity 65%
Wind Speed 12 km/h
Forecast Temperature 41°C
Risk Level High
Alert Status Active
Monitoring Date 16/9/2026
2.0°C difference from threshold
Early Warning: Issue Heatwave Warning


Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
ADVISORY ENGINE
Early Warning
Heatwave Risk and Early-Warning Decision
Evaluate temperature, humidity and forecast conditions to generate an early-warning recommendation.
Risk Assessment
Location
Pune
Current Temperature (°C)
39
Humidity (%)
65
Forecast Temperature (°C)
41
Prioritize vulnerable population alerts
Generate Advisory
Risk Level
HIGH
Pune
Temperature
39°C
Humidity
65%
Forecast Temperature
41°C
Recommendation
Issue Heatwave Warning and prioritize vulnerable population alerts


Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
LIVE MONITORING
Temperature Monitor
Heatwave Temperature Analysis
Compare the current temperature with the heatwave threshold and review the monitoring status.
Current Weather Observation
Location
Mumbai
Monitoring Date
16-09-2026
Current Temperature (°C)
39
Heatwave Threshold (°C)
37
Humidity (%)
65
Check Heatwave Status
Open Status Summary
Heatwave Condition Detected
Location
Mumbai
Current Temperature
39°C
Heatwave Threshold
37°C
Temperature Difference
2.0°C
Humidity
65%
Monitoring Date
16/9/2026


Home
Farm Dashboard
Irrigation Advisory
Heatwave Monitoring
Sensor Data
Contact
Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
ALERT SUBSCRIPTION
Heatwave Alerts
Develop a Climate Intelligence Registration and Early-Warning Form
Register for localized heatwave monitoring and choose how early-warning alerts should be delivered.
Climate Intelligence Registration
User Name *
Kehan Desai
Mobile Number *
9876543210
Email *
name@example.com
User ID *
CLM-1234
Location / City *
Mumbai
PIN / Postal Code *
400077
Age Group *
Select age group
User Category *
Select category
Preferred Alert Channel *
Select alert channel
Registration Date *
16-09-2026
Register for Heatwave Alerts
Clear Form

Quick Links
Weather Overview
Heatwave Monitoring
Early-Warning Advisory
Temperature Readings
Alert Registration
ALERT SUBSCRIPTION
Heatwave Alerts
Develop a Climate Intelligence Registration and Early-Warning Form
Register for localized heatwave monitoring and choose how early-warning alerts should be delivered.
Climate Intelligence Registration
User Name *
Kehan Desai
Use alphabets and spaces only.
Mobile Number *
9876543210
Enter a valid 10-digit Indian mobile number.
Email *
name@example.com
Enter a valid email address.
User ID *
CLM-1234
User ID must be in CLM-1234 format.
Location / City *
Mumbai
Enter a valid city or location.
PIN / Postal Code *
400077
PIN must contain exactly 6 digits.
Age Group *
Select age group
Select an age group.
User Category *
Select category
Select a user category.
Preferred Alert Channel *
Select alert channel
Select an alert channel.
Registration Date *
16-09-2026
Register for Heatwave Alerts
Clear Form



Conclusion and Discussion:

In this experiment, I developed a JavaScript-based Climate Intelligence system for heatwave monitoring and early warning. I used variables, operators, conditional statements, objects, arrays, functions and mathematical calculations to analyze temperature, humidity and forecast conditions and determine the appropriate heatwave risk level. I also used Regular Expressions and DOM manipulation to validate user registration details, display suitable error messages and prevent invalid form submission. Overall, the experiment demonstrated how JavaScript can be used to make a Climate Intelligence website interactive, perform climate-data analysis and provide useful heatwave alerts and early-warning recommendations.